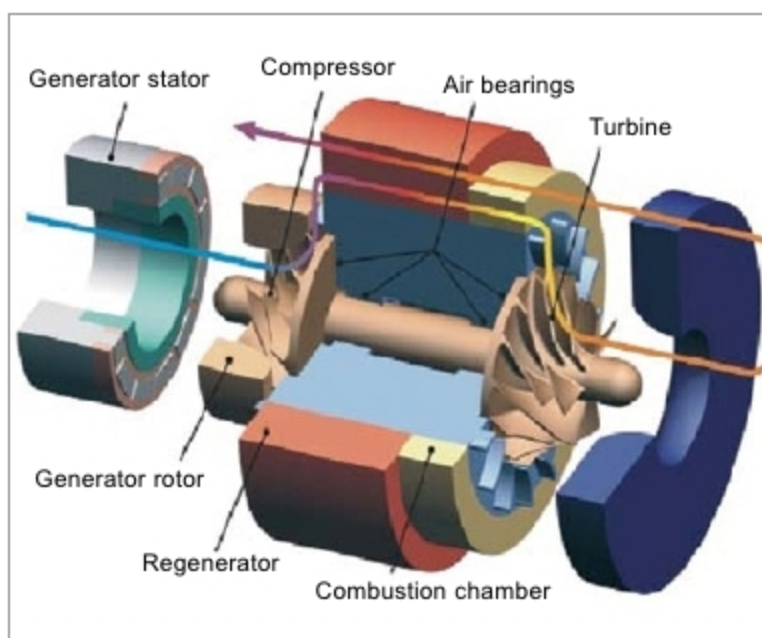


backup gen-set applications, where we today see diesel generator (DG) sets.”

Their first product is a compact and lightweight MGT for 20kVA generator. The generator is run by a single-shaft MGT engine. It does not require cooling fluid or oil. For higher power, these turbines can be used in series. Its minimum life is 10,000 hours, or ten years. Manufacturing began in December 2018. Grover says, “Currently, we have fuel-agnostic 20kVA and 100kVA units. Using these two systems, we can provide the whole portfolio of power from 20kVA to 1000kVA. Efficiency is 43 per cent electric even at twenty per cent partial loads, making it on average fifty per cent more efficient than a DG set.”

Being fuel-flexible, MGTs can run on both liquid fuels, like biodiesel and bioethanol, as well as gaseous fuels, like biogas and natural gas. Hence, biowaste from sugar mills and textile factories, among others, can be used as source, which decreases cost.

The company’s patented technology reduces both nitrogen-oxides (NOx) and other emissions (like particulate matter) to a negligible level, and can burn very low calorific value fuels as well, which makes the



generator environmentally friendly. Grover adds, “Pollution levels are hundred times less than current emission norms, that is, CPCB II (Central Pollution Control Board).”

MGTs can run for long durations, and maintenance cost is low as frequent servicing is not required. All these factors make these useful as a reliable solution for off-grid applications. These can serve as backup power for several places like offices, apartments, healthcare centres and data centres. As an alternative to DG sets, MGTs find applications in oil rigs, heating and cooling systems. Electric vehicles can also use these as onboard

chargers to power batteries.

The company plans to market the product at a low price to reduce the need for imports and make it affordable to the masses, thereby boosting Make in India initiative. Regarding support, Grover explains, “We received a grant from Bharat Petroleum Corp. Ltd (BPCL) in 2018 as part of their startup support initiative—Project Ankur—which was under the government’s Startup India

scheme. Other than that, we have received a lot of support from IIT-Madras labs, mainly National Centre for Combustion Research and Development.”

Discussing the startup’s future plans, Grover says, “Our prospective clients who depend on diesel fuel for power generation can shift to liquefied petroleum gas (LPG), which is cheaper and cleaner than diesel. Other than replacing the IC engine in the DG sets, our next target is to work on doing the same for four-wheelers. Ashok Leyland has shown interest to use these in its intercity buses.”

—Ayushee Sharma

4 3D PRINTER FOR HOBBYISTS AND SMALL INDUSTRIES

The process of making a three-dimensional (3D) solid object from a digital file is known as 3D printing or additive manufacturing. A 3D object is created through an additive process, which means laying down successive layers of material until the object is created. Additive manufacturing is the opposite of subtractive manufacturing, which is done through cutting out or hollowing out a piece of metal or plastic to achieve the desired shape.

3D printing enables producing complex shapes using less material than traditional manufacturing methods. It starts with a 3D model on a 3D modelling software—users can create one themselves, or download it from a 3D repository. There are four major

types of materials used for 3D printing, namely, metal, polymer powder, resin and plastics like PLA, ABS, nylon, PP, etc.

MESH3 is a rapid hardware prototyping platform with capabilities ranging from 3D designing and printing to embedded systems and PCB design.

MESH3 has recently released NiK-0 3D printer, which has been successfully adapted and engineered keeping in mind functional and scalable aspects. This printer provides printing reliability and ease of use at only ₹ 37,500.

Keeping up with the hobbyist ideology, the printer is engineered to be hassle-free and completely automatic. It features an inductive sensor, which means the users will never have to worry about levelling the print bed



(Credit: www.mesh3.in)

ever again. Moreover, an unconventionally large build volume of 220mm x 220mm x 300mm gives them the freedom to print what they love. This